

Chapter 3: Cellular Respiration

Cellular respiration is the process by which cells convert nutrients into ATP, the energy currency of the cell.

Glycolysis is defined as the breakdown of glucose into pyruvate, occurring in the cytoplasm.

Why does the process require oxygen in later stages?
Because the electron transport chain needs oxygen as the final electron acceptor, without which ATP production would halt almost entirely.

Key Terms

Mitochondria: the organelle known as the powerhouse of the cell.

ATP: adenosine triphosphate, the primary energy carrier.

Krebs cycle: also known as the citric acid cycle.

NADH: an electron carrier produced during glycolysis.

3.2 Comparing Aerobic and Anaerobic Respiration

Aerobic respiration produces far more ATP per glucose molecule than anaerobic respiration, however it requires a constant oxygen supply.

For example, muscle cells switch to anaerobic respiration during intense exercise, producing lactic acid as a byproduct. This trade-off between speed and yield illustrates the relationship between oxygen availability and energy production efficiency.